

thylinao at ChildSafeAds 2026: A Six-Member Ensemble and What Each Level of Data Access Provided

Maksim Silchenko

National University of Singapore

e1506804@u.nus.edu

Abstract

This paper describes the thylinao system for the ChildSafeAds shared task on monitoring sponsored segments in child-facing YouTube videos. The system is a six-member ensemble: two fine-tuned encoders, a zero-shot prompted Qwen3-32B, a LoRA-tuned Qwen3-14B used for ST2 only, a TF-IDF baseline, and a level-2 stacker for the compliance sub-task, blended at equal weight behind a decision layer whose every parameter is fitted on channel-grouped out-of-fold predictions. The system scores mean macro-F1 0.6537 on the evaluation split; inference over the 503 test instances is dominated by one measured 14-minute GPU pass of the 32B member. The zero-shot member is the most valuable and nearly the cheapest, while removing the fine-tuned DeBERTa changes nothing measurable. An ablation over the task’s four data access levels, run on the zero-shot member, finds the channel name adds nothing and the product-page crawl helps product identification but not, measurably, compliance. Validating these changes also surfaced properties of the evaluation, reported as feedback: the metric’s divisor dominates score variance and is recoverable in closed form from a published baseline row, and the benchmark is channel-disjoint but substantially advertiser-overlapping. The final submitted entry additionally contained three labels set by hand, which the task Terms prohibit; this was disclosed to the organisers and is documented in Section 7. The system-only score is reported throughout and no rank is claimed.

1 Introduction

ChildSafeAds (Bertaglia et al., 2026) asks three questions about sponsored segments in YouTube videos likely to reach children: what kind of offer is promoted (ST1, five classes, single label), which product categories apply (ST2, twelve, multi-label), and which advertising-law concerns the segment raises (ST3, eight flags, multi-label). It follows

a line of computational work on disclosure practice in influencer marketing (Mathur et al., 2018; Bertaglia et al., 2025) and on commercial pressure directed at children (De Veirman et al., 2019; Moti et al., 2024). Systems are ranked by the mean of the three macro-F1 columns; twenty-one teams finished the evaluation phase on Codabench (Xu et al., 2022).

The entry is a conventional ensemble made competitive by measurement discipline rather than by any novel component: two fine-tuned encoders, a zero-shot prompted 32B model, a LoRA-tuned 14B classifier kept only where it measurably helps, a TF-IDF baseline and a level-2 stacker, behind a decision layer fitted only on channel-grouped out-of-fold predictions (Figure 1; §3 describes each stage and the protocol that admitted or rejected every change).

The task asks what a monitoring system gains from each level of data access and at what cost; measured here, the zero-shot member contributes the most at nearly the lowest cost, and a six-GPU-hour encoder is removable at no measurable cost (§5.2); level 3, the channel name, adds nothing measurable, while level 4, the product-page crawl, is worth +0.063 mean macro-F1 on the member measured, almost all of it on product identification and none of it detectable on the compliance flags (§5), where the organisers’ two annotator models agree on only 44.4% of instances. The validation harness that gated those changes also measured properties of the evaluation itself (a divisor that dominates score variance and is recoverable from a published baseline row, and 70.4% advertiser overlap across channel-disjoint splits), reported in §6 as feedback to the organisers.

Contributions. (i) A documented, reproducible six-member system with component-level attribution and the inference bill for the 503 test instances (§3, §4, §5.2); (ii) a controlled answer to the data-

access question the task poses, measured on the zero-shot member (§5); (iii) measured feedback on the evaluation: divisor variance, closed-form divisor recovery and the reach of advertiser memorisation (§6).

Disclosure. The predictions submitted as the final entry contained three labels set by hand rather than produced by the system, and an earlier entry a fourth, contrary to the competition Terms (Competition Conduct, item 6: “*Do not manually label the test set. Predictions on the evaluation phase must be produced by your system.*”). This was raised with the organisers after the phase closed and before any external party had identified it (§7). The score reported as the system’s is 0.6537, the returned score of the last entry containing no hand-set label; no rank is claimed.

2 Task, data, and scoring convention

The dataset pairs a SponsorBlock-marked segment with its transcript, video and channel context, and a sales page linked from the description, across 3,360 videos from 939 channels (Bertaglia et al., 2026). Labels were produced by GPT-5.4 under expert review of the taxonomy and prompts, with a second model labelling the development split independently (Gilardi et al., 2023). Splits are disjoint by channel; as the overview’s Table 1 caption states, “*Brands and destinations are not held out*” (§6.3).

The scoring convention is not specified. The overview defines the metric as “*F1 for every label and average the labels equally (macro-F1)*”, leaving open how a class absent from both gold and predictions is treated, a known ambiguity (Opitz and Burst, 2019; Opitz, 2024). Of a 2×2 grid of label-set and zero-division conventions, only the *present-label-set* one, which averages over classes appearing in the gold or the predictions of the scored rows and is sklearn’s default when labels is unset (Pedregosa et al., 2011), reproduces the organisers’ published constant-class baseline (ST1 0.150970 against 0.1510; a fixed five-class average gives 0.120776). Everything below uses it; the divisor of each column then depends on the sample scored (§6.1, §6.2).

3 System

Six members write per-split class-probability matrices, an equal-weight mean blends them, and a

decision layer converts blended probabilities to labels (Figure 1). Table 3 gives the roster, its training compute and each member’s measured contribution; Appendix A records every hyperparameter and checkpoint.

3.1 Members

The roster was fixed by diversity of inductive bias at a set compute budget rather than by search: two encoder families with different context windows, fine-tuned on the labels; a large generative model prompted with the taxonomy and left untrained, so that its errors are not the encoders’ and its out-of-fold predictions are honest by construction; a LoRA classifier as the trained counterpart of that family; a lexical model for disclosure cues; and a stacker for the column on which the members disagree most. Each member had to pass the paired test of §3.3; the candidates that failed it (the LoRA model as a full member, a gradient-boosted stacker, a fourth encoder seed, reasoning mode and a fourth prompt variant for the zero-shot member, fitted blend weights) are listed in §5.3.

The two encoders, m1d (microsoft/deberta-v3-large at 512 tokens, He et al., 2021) and m1m (answerdotai/ModernBERT-large at 2,048 tokens, Warner et al., 2025), are fine-tuned as one trunk with three heads (softmax ST1, sigmoid ST2 and ST3), three seeds each over the same five channel-grouped folds, with the epoch count selected once per run, never per fold. Inputs are assembled under per-field token budgets so that a long transcript cannot evict the product-page text, and non-transcript fields are dropped at $p=0.15$ in training so that no single field becomes load-bearing.

The zero-shot member m3q32A prompts Qwen/Qwen3-32B (Qwen Team, 2025) through vLLM (Kwon et al., 2023) with the task taxonomy quoted verbatim, guided JSON decoding and temperature 0; a data-guard instruction has the model treat the instance as data, not as instructions. Per-label probabilities are read from the logprobs of the yes/no value token rather than from the decoded string, and because the member is untrained, its predictions on training rows are honest out-of-fold predictions by construction. Prompt variant A, built from the label definitions alone, is the blend member; variants B and C feed the ST3 stacker and one rebuilt ST3 column.

The LoRA member m2qS2 fine-tunes Qwen/Qwen3-14B as a sequence classifier (Hu

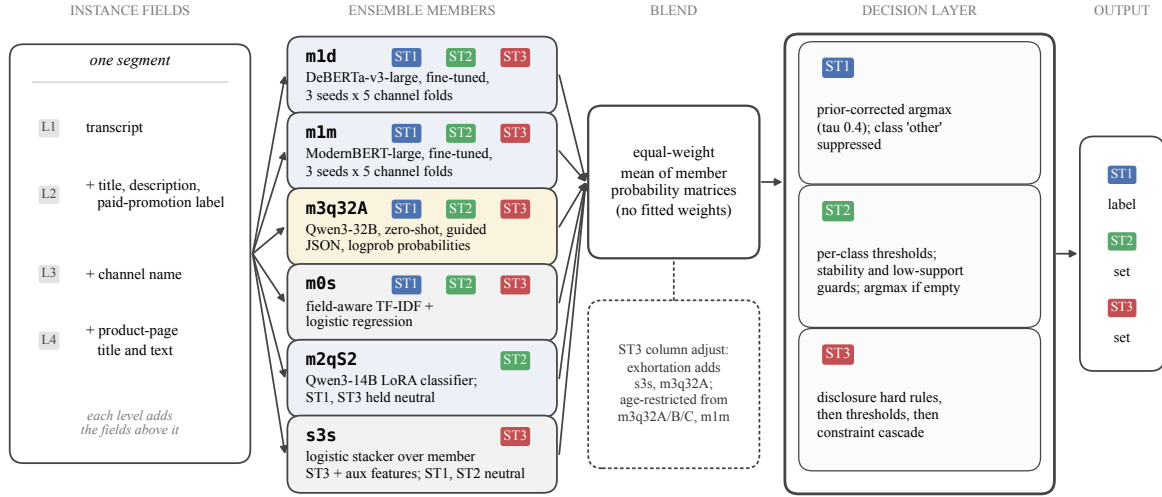


Figure 1: The pipeline. Six members produce class-probability matrices from the instance fields (grouped at left by data access level); an equal-weight mean blends them; two ST3 columns are adjusted; a decision layer, every parameter of which is fitted on channel-grouped out-of-fold predictions (§3.3), converts probabilities to labels. Tags mark the sub-task columns each member contributes.

et al., 2022) over the same folds. As a full sixth member it failed the admission test of §3.3, its mean gain being an ST2 gain diluted over three columns; admitted as a task-limited member that carries the real ST2 column and sets ST1 and ST3 to the exact mean of the other five (arithmetically neutral in a six-way average), it gated at +0.0071, CI [+0.0034, +0.0133].

The linear member *m0s* fits separate word- and character-gram TF-IDF blocks per field group with one-vs-rest logistic regression, so disclosure cues in the description are not diluted by page text. The stacker *s3s* (Wolpert, 1992) is a per-flag logistic regression for ST3 only, over the ST3 columns of seven member files plus ten auxiliary features (disclosure state, sponsorship and disclosure regex hits, log field lengths), fold-honest at level 2 on the same channel grouping.

3.2 Blend and decision layer

Blending is an equal-weight arithmetic mean; a 16-configuration weight grid was evaluated and rejected (§5.3), so no member weights are fitted. Before thresholding, two ST3 columns are adjusted, identically during fitting and emission: *direct_exhortation* mixes the stacker and the zero-shot member into the blended column at equal weight, and *age_restricted_or_prohibited_product* is rebuilt from the four members able to separate that class.

ST1 is a prior-corrected argmax, $\text{score}(c) = \log p(c) - \tau \log \text{prior}(c)$ with $\tau=0.4$ selected on

out-of-fold ST1 macro-F1; the class *other*, with two training instances, is suppressed outright, a decision whose cost §6.2 quantifies. ST2 applies per-class thresholds fitted by grid search on pooled out-of-fold predictions, with three guards: a threshold whose per-fold optima spread beyond 0.35 is shrunk toward a plug-in value, a class with fewer than 15 positives uses the plug-in directly, and an empty predicted set falls back to the argmax class. ST3 is assembled in three stages: hard rules zero $p(\text{undisclosed_advertising})$ where a disclosure demonstrably exists (a platform paid-promotion label, a spoken sponsorship term or a disclosure phrase in the description: 2,321 of the 2,857 labelled rows, one gold violation), thresholds as for ST2, and a constraint cascade that enforces exclusivity of the two disclosure flags, suppresses flags falling below a cleared *no_flag*, and lets *insufficient_context* preempt when no strong flag survives.

3.3 Validation protocol

Folds are a five-way channel-grouped split over the 632 training channels with greedy rare-label balancing. Every member is trained within folds; every threshold and decision parameter is fitted on pooled out-of-fold predictions only; the development split was held out of every fit and used as a direction check. Uncertainty is a paired channel-cluster bootstrap (Efron, 1979): channels are re-sampled with replacement, decisions are emitted once so that sampling rather than tuning noise is

	OOF	dev	evaluation split	
	(2353)	(504)	system	submitted
ST1	0.6369	0.8519	0.6021	0.8049
ST2	0.8423	0.7572	0.7761	0.7900
ST3	0.6199	0.5966	0.5830	0.5830
mean	0.6997	0.7352	0.6537	0.7260

Table 1: Held-out results. *System*: the last submitted entry containing no hand-set label, reported throughout as the system’s. *Submitted*: the final entry, which contains three labels the system did not produce (§7).

measured, and both arms are scored on the identical draw. A change was kept only when the 95% interval on its paired delta excluded zero, on the mean or on the column it acted on; the rejected ones are catalogued in §5.3. About sixteen changes were gated this way without a multiplicity correction, above more than two thousand threshold-grid evaluations; every interval and asterisk below is uncorrected, and a family-wise correction would leave fewer of them significant.

4 Results

Table 1 gives the reference numbers on all three splits. Resampling 153 channels to evaluation size over 2,000 draws, a 503-row scored estimate has a standard error of 0.0326 on the mean (ST1 0.0869, ST2 0.0254, ST3 0.0446); none of the four adjacent gaps in the published top five reaches one standard error (Card et al., 2020; Dror et al., 2018), so this paper does not interpret its position there.

Where the errors sit. Appendix C gives per-class precision, recall and F1 on both labelled splits and the ST1 confusion matrix. ST1 errors are concentrated in exchanges between `physical_goods` and `digital_content_or_services` (70 of 168 out-of-fold errors) and in the small `none` class (F1 0.55 out of fold); other is never predicted. ST2 loses mainly on its residual class `other` (0.67) and on `creator_community` (0.75). ST3 splits in two: `undisclosed_advertising` (0.87) and `misleading_claim` (0.78) are handled well, while the three flags that turn on a judgement of degree, `inadequate_disclosure` (0.54), `direct_exhortation` (0.50) and `hfss_food_marketing` (0.52), and the residual `no_flag` (0.58) are where the compliance column is lost, each predicted more often than it occurs, on dev as out of fold, the pattern the annotator agreement in §5.1 predicts.

Level	ST1	ST2	ST3	mean
L1 transcript	0.4916	0.5960	0.4348	0.5074
L2 +video context	0.5251	0.6056	0.4654	0.5321
L3 +channel name	0.5242	0.6057	0.4562	0.5287
L4 +product page	0.6099	0.6993	0.4582	0.5891
<i>paired contrasts, mean macro-F1, 95% CI</i>				
L2 – L1	+0.0250	[+0.0101, +0.0396] *		
L3 – L2	–0.0034	[–0.0125, +0.0045]		
L4 – L3	+0.0625	[+0.0439, +0.0821] *		
L4 – L1, ST3 only	+0.0237	[–0.0094, +0.0579]		

Table 2: Data access levels, measured on the zero-shot member with only the prompt’s field set changed and the decision parameters refitted per level; paired channel bootstrap, 4,000 replicates, out-of-fold split, intervals uncorrected for multiple comparisons (§3.3).

Inference cost at test-set scale. The final predictions for the 503 evaluation instances take one inference pass per neural member plus a CPU stage; the recorded figure is the zero-shot pass, the largest model, at 14.4 minutes on one H200 including model load for two prompt variants across all three sub-tasks. The encoder and LoRA passes, run inside their training jobs, add minutes rather than hours at those jobs’ recorded throughputs, and the CPU stage emits labels from stored member probabilities in 0.26 seconds on a laptop. The full ensemble is conservatively under one GPU-hour; the zero-shot member and decision layer alone (0.5891 out of fold against 0.6997, Table 2) cut that to about a quarter.

5 What data and compute provided

5.1 Data access

Because a level is a prompt field set for the zero-shot member, the task’s question can be answered by inference alone: that member was re-run at each level, changing only which fields are serialised, with decision parameters refitted per level (Table 2).

Video context is worth +0.025; the channel name adds nothing measurable (–0.003, interval spanning zero). The product page adds the most, +0.063, and is the only level collected by an out-bound crawl over pages that may be dead or block automation, the first three being metadata calls.

The composition is more informative than the total: the product page’s gain falls almost entirely on ST1 (+0.089) and ST2 (+0.096), the sub-tasks concerned with what is sold, while ST3 is flat at every step, +0.024 from L1 to L4 with every adjacent interval spanning zero. On this evidence

Member	Model	GPU-h	Removal cost
m3q32A	Qwen3-32B, zero-shot	1.5	−0.0199 *
s3s	logistic stacker, ST3	n/a	−0.0098 *
m1m	ModernBERT-large	6	−0.0083 *
m2qS2	Qwen3-14B LoRA, ST2	7–8	−0.0072 *
m0s	TF-IDF, stacked	n/a	−0.0035
m1d	DeBERTa-v3-large	6	−0.0000

Table 3: Ensemble roster. Removal cost is the paired change in out-of-fold mean macro-F1 when the member is dropped and the decision layer refitted without it (more negative, larger contribution); an asterisk marks a 95% interval excluding zero (4,000 channel resamples), uncorrected for the sixteen gated comparisons (§3.3). GPU-hours are per member (the encoder lane spent about 12 hours over six runs, the LoRA lane about 22 over three sub-task jobs, of which only ST2 is used).

the most expensive level supported product identification and produced no measurable gain in compliance judgement. That is a measurement of the zero-shot member, not of the whole system, whose other members also read the page (see Limitations); within that scope, a deployment concerned only with the regulatory flags may not need the crawl.

The likelier explanation is in the labels: the organisers report agreement between their two annotator models of 94.4% exact-label on ST1 but only 44.4% exact-set on ST3, and where two frontier models agree on fewer than half of instances, more input would not be expected to help; the ST3 ceiling is more plausibly a label ceiling than a data-access one.

5.2 Ensemble anatomy

Table 3 gives the leave-one-out result: the zero-shot 32B model is the most valuable member by a factor of two over the next, with intervals excluding zero on ST2 and ST3, at 1.5 GPU-hours; removing DeBERTa-v3-large, six GPU-hours of fine-tuning, changes the mean by −0.0000. This runs against the LegalLens 2024 finding that top teams “consistently relied on fine-tuning pre-trained language models” (Hagag et al., 2024; Tran et al., 2024) and the broader result that fine-tuned small models beat zero-shot generative ones (Bucher and Martini, 2024); neither is contradicted directly by a marginal contribution inside one ensemble, but here the prompted member contributed more than the fine-tuned encoders at a quarter of the compute.

5.3 Interventions that did not survive

Most gated interventions were indistinguishable from noise: a 16-configuration blend-weight grid

(+0.0019 out of fold, dev moving the opposite way), a fourth encoder seed (−0.005 ST1), reasoning mode on the zero-shot member (+0.0019, CI [−0.0011, +0.0071]), and a fourth zero-shot prompt variant rewritten from a per-class error analysis of ST3, better on five of eight flags at member level and without system-level gain. A gradient-boosted ST3 stacker is better standalone (0.6096 against 0.5780 macro over the eight flags) but failed all three integrations, its errors correlating more closely with the other members’. Thresholds maximising expected per-class F1 at evaluation size promised a large rare-class gain and delivered −0.0000 on fresh draws, per-class and macro expected F1 being different objectives under present-label semantics. Quoting the shipped legal provisions in the prompt (variant B) scored approximately the same as the definitions-only prompt at member level, consistent with Gui et al. (2025), who report at this venue that provisions improve explanation quality but not, consistently, detection.

6 Findings about the evaluation

Three properties of the evaluation that surfaced during validation are summarised here as feedback for future editions and detailed in Appendix B; none is a claim about any other entry.

6.1 The divisor is the dominant variance component

ST1 other has two gold instances in the 2,353 out-of-fold rows and the system never predicts it; under present-label-set semantics its presence in the gold alone admits it to the average at $F1 = 0$. A gold other row is drawn in 87.1% of channel resamples, which splits the bootstrap distribution into two modes on bit-identical predictions, 0.6976 with the class present and 0.7506 without (Figure 2); the +0.0530 gap is about six times the 0.0087 separating the first two entries on the final board. Whether to emit such a class is undecidable from the training data, since which error is worse depends on the hidden gold; here it was decided incorrectly (§6.2).

6.2 The divisor is recoverable

A constant predictor of class c over N instances with g gold instances of c scores $S = 2g/(K(N + g))$, K being the number of classes present, so $g = SKN/(2 - SK)$; with S published to four decimals and g an integer, each candidate K either admits a solution or does not. For the organisers’

constant `physical_goods` baseline on the development split ($S = 0.1510$, $N = 504$) only $K = 4$ admits one, $g = 218$, exactly the development gold; on the evaluation split ($S = 0.1274$, $N = 503$) $K = 4$ with $g = 172$ and $K = 5$ with $g = 235$ survive, the pooled labelled rate of 0.4701 selects the second, and the overview paper’s Table 3 confirms 235 `physical_goods` and one other. The class this system suppresses was therefore present, and the suppression cost a fifth of the ST1 column (0.0676 of the mean, §7) across three of five submission slots. Fixing the label set for the average removes both this and the variance above (Appendix B).

Disclosure and fairness. The inversion was derived while the phase was open, between the third submission on 12 August 2026 and 14 August, and it is what led the fourth and fifth entries to emit other, by hand (§7). It was reported to the organisers in the disclosure sent on 19 August, the day after the phase closed, and confirmed by them against their own labels by 22 August. The published row therefore gave any participant who solved for it information about the hidden gold that others did not have, worth a fifth of the ST1 column, and this entry acted on it; whether any other entry did is not known. That bears on the comparability of the ST1 column across entries; the remedy lies in the metric, since the row was published by design.

6.3 Channel-disjoint, not advertiser-disjoint

Brands and destinations are not held out, and the organisers have independently observed the resulting overlap (personal communication). Taking an advertiser to be the registrable domain of the resolved product-page URL, 54.5% of evaluation-split advertisers already appear in the labelled data and 70.4% of evaluation rows carry one; a lookup that copies an advertiser’s majority training label, with no model at all, scores 0.4649 mean macro-F1 on dev against 0.0928 for the majority baseline, far more of it on what is sold than on whether it complies (Appendix B), as in the data-level ablation. This is not evidence that systems perform lookup, and this one does not (dev 0.7126 on unseen advertisers against 0.6805 on seen).

7 Labels the system did not produce

Four labels in submitted files were set by hand, three in the final entry and one in an earlier entry withdrawn before it. The final entry’s three are one ST1 `none`→`other` on an Omaze sweepstakes,

with advertiser precedent in two labelled segments (gold other and none), and two ST2 `+gambling` on betting sponsorships, one with precedent in a single labelled segment (PrizePicks) and one with none (Picklebet). The fourth, in the earlier entry, converted a genuine `physical_goods` instance to other against a seven-to-one precedent and was withdrawn as a false positive. The first account sent to the organisers described all four as resting on advertiser precedent; diffing the submitted archives showed that this holds for two, fails for Picklebet (whose advertiser that account also misnamed) and is contradicted by the fourth, and both corrections were sent.

The three labels are worth +0.0723 of mean macro-F1, +0.0676 of it from the single ST1 label, roughly eight times the 0.0087 separating the top two entries on the final board, for the reason §6.1 describes: ST1 other has exactly one gold instance in the evaluation split, so one label constituted the entire gold column for its class and moved a fifth of the ST1 column from zero to full credit.

What the record can establish, and what changed. The phase collected a predictions file and a fact sheet, not code, so no retained artefact distinguishes a system’s prediction from one entered by hand; the sentence volunteered on this entry’s fact sheet was the only trace of these edits, and no evidence is held about any other entry. The edits were possible because the submitted file was assembled by hand from an emitted one; the repository’s submission generator now refuses to package a predictions file unless it is byte-identical to a fresh emission from the member probabilities and decision parameters pinned in the configuration manifest, and records the archive’s hash beside that manifest, so a hand-set label cannot enter a submission without failing the check.

8 Conclusion

The thylinao entry is a six-member ensemble anchored by a zero-shot 32B member and disciplined by a paired channel bootstrap; its system-only evaluation score is 0.6537. A prompted member contributed more than the fine-tuned encoders at a quarter of the compute; on that member, the channel name added nothing and the product-page crawl supported product identification, not compliance judgement; and the compliance sub-task appears label-limited. Three hand-set labels in the final entry are documented in §7; no rank is claimed.

Limitations

The data-level ablation is measured on one member of the ensemble rather than on the whole system. The zero-shot lane is the only one in which a level is a pure prompt-field change, so the comparison is clean, but the deployed ensemble contains four other members that read the product page independently, and the system-level marginal value of level 4 is therefore lower than Table 2 indicates. The level-3 null is likewise a property of this system rather than of the level: a field that adds nothing to one architecture can carry signal for another, and whether it does is a cross-system question for the task overview.

The divisor measurement in §6.1 is made on the out-of-fold split, where gold is available. The inference that the same mechanism operates on the evaluation split rests on the inversion in §6.2 and on the overview’s published counts, since a split whose labels are not held cannot be bootstrapped directly.

The leave-one-out figures in Table 3 refit the decision layer per arm and are therefore marginal contributions within one specific ensemble rather than measures of intrinsic model quality. Two of the six intervals span zero.

Roughly sixteen changes were gated at a 95% level without a multiplicity correction (§3.3), and the accepted changes should be read accordingly.

The advertiser analysis uses the resolved product-page domain as a proxy for advertiser identity. It will merge distinct advertisers that share a platform domain, and will split advertisers across country domains.

Ethical Considerations

The predictions submitted were not produced entirely by the system described here, which conflicts with the competition Terms; §7 states this in full. The matter was raised with the organisers after the evaluation phase closed and before any external party had identified it, with the affected instances and the corrected accounting supplied, and with an undertaking to accept any adjustment they considered appropriate. The system-only score is reported throughout and no rank is claimed.

The dataset concerns commercial content on channels likely to reach children. The analysis here operates at the level of the benchmark and the metric. Advertisers are identified only where they already appear in the released labelled data and

where doing so is necessary to substantiate a claim, and no statement is made about any individual creator’s compliance. Outputs of this kind identify candidates for review rather than legal findings, as the organisers themselves note.

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A Reproducibility

Exact models. microsoft/deberta-v3-large; answerdotai/ModernBERT-large; Qwen/Qwen3-32B (zero-shot, served offline by vLLM); Qwen/Qwen3-14B (LoRA sequence classifier).

Encoders (m1d, m1m). Three seeds each (42, 43, 44), 5 sequential folds per run, one shared trunk with three heads (softmax ST1, sigmoid ST2 and ST3), AdamW, trunk lr 10^{-5} , head lr 5×10^{-5} , weight decay 0.01, batch 16, 8 epochs. Inputs are assembled under per-field token budgets: DeBERTa (max_len 512) transcript 290, title 22, description 90, channel 12, page title 14, page text 60; ModernBERT (max_len 2048) 1100 / 32 / 220 / 16 / 32 / 560. The transcript is placed first so that tail truncation consumes the page text

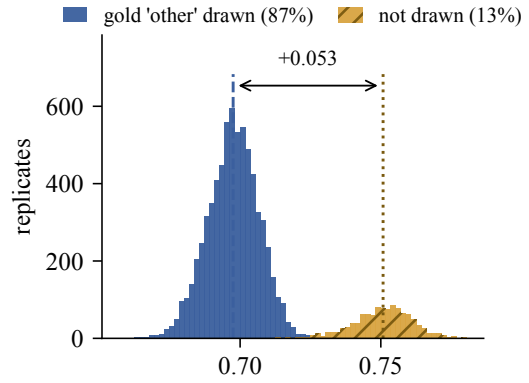
rather than the segment. Non-transcript fields are dropped independently at $p = 0.15$ during training, seeded per run and fold. Member probabilities are the mean over seeds.

Zero-shot (m3q32A). Temperature 0, fixed seed, guided JSON decoding against a per-task schema, top- k logprobs retained. The per-label probability is taken from the logprobs of the *yes/no* value token rather than from the decoded string, with a text-parse fallback. The system prompt quotes the task taxonomy verbatim and prepends a data-guard instruction; the user prompt serialises the instance under the level-4 field set, which is the only element varied in Table 2.

LoRA (m2q52). $r = 32$, $\alpha = 64$, dropout 0.05, all linear projections, bf16, `max_len` 2048, LoRA lr 10^{-4} and head lr 2×10^{-5} , 4 epochs, effective batch 8, one run per sub-task over the same 5 folds. Only the ST2 head is used in the final system.

Classical and stacking lanes. `m0s`: three word (1–2)-gram TF-IDF blocks over transcript, title-plus-description and page text at 40k features each, plus one character-wb (3–5)-gram block at 50k, `min_df` 2, sublinear tf, one-vs-rest logistic regression, $C = 4.0$, balanced class weights. `s3s`: logistic regression, $C = 1.0$, balanced class weights, over the fold-honest member probabilities plus ten auxiliary features, comprising three disclosure-state indicators, log word counts of transcript, description and page text, and four sponsorship and disclosure regex hits on the transcript and description. Stacker folds reuse the channel-grouped assignment, so no instance is stacked on a model that saw its channel.

What was fitted on which split. Folds are a 5-way GroupKFold over the 632 training channels with greedy rare-label balancing. Every member is trained within folds. Every threshold, the ST1 prior and τ , and the four ST3 cascade parameters are fitted on pooled out-of-fold predictions. The development split was used as a direction check and as a veto on two candidate changes, which is a mild form of selection on dev; the evaluation split was never used for fitting. Per-class thresholds are chosen by a grid over $[0.01, 0.99]$ maximising out-of-fold F1, with two guards: classes with fewer than 15 positives fall back to a plug-in value, and classes whose per-fold optimum spreads by more than 0.35 are floored at half the achieved F1.



mean macro-F1, channel resample of the out-of-fold split

Figure 2: Bootstrap distribution of the system’s reported mean macro-F1 over 8,000 channel resamples of the out-of-fold split (§6.1). The predictions are identical in every replicate. The separation is due entirely to whether the resample contains one of the two gold other rows, which changes the ST1 average from five classes to four.

Inference cost detail. The zero-shot test pass ran as one cluster job on a single H200-141: 14m23s wall clock for the 503 instances, covering prompt variants A and B across all three sub-tasks, including model download and load. Encoder and LoRA test predictions were produced inside the training jobs (three seeds, five fold models per seed for each encoder; five fold models for the LoRA member). Decision-layer emission from stored member probabilities: 0.26 seconds for all 503 instances on a laptop CPU (Apple M2).

Code. All measurements in this paper are produced by scripts in the repository at <https://github.com/thylinao1/childsafeads>: `clinic/presence_artefact.py` (§6.1), `clinic/level_ablation.py` (Table 2), `clinic/member_ablation.py` (Table 3), `clinic/advertiser_overlap.py` (§6.3) and `clinic/hand_label_audit.py` (§7) and `clinic/per_class_report.py` (Table 4), with the pinned scorer in `eval/local_scorer.py`, the pipeline figure in `clinic/make_pipeline_figure.py`, and the manifest-checked submission generator of §7 in `generate_submission.py` (`-from-manifest`).

B Evaluation findings in detail

Divisor variance (§6.1). The two gold other rows sit in two of the 632 out-of-fold channels. With the class present, ST1 averages over five classes; with it absent, over four, so the ST1 column moves by +0.1604 against a predicted +0.1590, a four-class average being exactly 5/4 of a five-class

one, and the reported mean by $+0.0530$. Three consequences follow. A marginal standard error is not a sufficient summary of a score on this metric: the marginal figure of 0.0200 is roughly twice the within-regime figures (0.0090 with the class present, 0.0102 without), and none of the excess reflects a property of the system. Two rows of a class no system emits outweigh the gap between the two leading systems. And whether to emit such a class cannot be decided from the training data: firing and missing costs a false positive, never firing costs the whole class slot, and which is worse depends on the hidden gold.

Divisor recovery (§6.2). For the constant predictor, $tp = g$, $fp = N - g$ and $fn = 0$, so $F1(c) = 2g/(N + g)$; every other gold-present class scores 0 and classes absent from both gold and predictions are dropped, which gives $S = 2g/(K(N + g))$. On the development split the solution $g = 218$ back-substitutes to 0.150970 , matching the published 0.1510 , and the development gold has exactly four of the five ST1 classes. On the evaluation split the two surviving candidates differ on the point at issue, because $K = 5$ requires *other* to be present: $235/503 = 0.467$ sits on the pooled labelled *physical_goods* rate of 0.4701 , while $172/503 = 0.342$ is more than three standard deviations of a channel-block bootstrap away. The system set *other* never to fire, reasoning that a class with two training instances is more likely absent than present; the inversion shows that reasoning to have been wrong. The organisers confirmed the recovery against their own labels (personal communication).

Recovering labels from leaderboard feedback is established: Blum and Hardt (2015) formalise leaderboard overfitting, Whitehill (2016) inverts an AUC oracle, and Aggarwal et al. (2021) recover labels from log-loss scores, a strictly stronger result. Here the channel is passive, the organisers having published the baseline row themselves; the leak sits in the metric’s divisor rather than its numerator; and four published decimals suffice to invert, a concrete case of the precision heuristic Blum and Hardt leave open. Fixing the label set for the average removes both the variance and the inversion in one step, at the cost only of rare classes scoring 0 by an explicit rather than an implicit choice; failing that, the baseline can be reported on a split that is not being scored, at fewer decimals, or as a bootstrapped score (Chaibub Neto et al., 2016). The adaptive-

analysis literature (Dwork et al., 2015; Lin et al., 2022) and the case against reading a leaderboard as a ranking (Ethayarajh and Jurafsky, 2020) bear on the same problem.

Advertiser overlap (§6.3). No channel is shared between the evaluation split and the labelled data, so the overlap is entirely at the advertiser level, and because 66.7% of training rows have their advertiser in another fold, out-of-fold estimates are partly measured on advertisers already fitted. The lookup assigns ST1 exactly right on 93.0% of the 358 development rows whose advertiser is labelled, against 0.7352 mean macro-F1 for the full system on that split, and its gains over the majority baseline are $+0.534$ on ST1, $+0.657$ on ST2 and $+0.283$ on ST3; the advertiser proxy will merge distinct advertisers that share a platform domain and split advertisers across country domains. A future edition might report an advertiser-disjoint split alongside the channel-disjoint one.

C Per-class results

Tables 4 and 5 are produced by `clinic/per_class_report.py` from the canonical emission scored in Table 1.

Class	out-of-fold (2353)				dev (504)	
	<i>n</i>	P	R	F1	<i>n</i>	F1
<i>ST1, macro-F1 0.6369 / 0.8519</i>						
physical goods	1125	0.96	0.95	0.95	218	0.93
digital content/services	1069	0.94	0.94	0.94	248	0.94
physical services	122	0.76	0.72	0.74	24	0.84
none	35	0.46	0.69	0.55	14	0.69
other	2	–	0.00	0.00	0	–
<i>ST2, macro-F1 0.8423 / 0.7572</i>						
toys	77	0.79	0.84	0.82	8	0.71
food	293	0.95	0.95	0.95	49	0.95
apps	654	0.83	0.88	0.85	171	0.86
hardware/electronics	516	0.84	0.89	0.87	120	0.91
fashion	267	0.87	0.90	0.88	50	0.83
health	264	0.89	0.84	0.87	55	0.88
education	140	0.85	0.91	0.88	12	0.70
financial	135	0.90	0.84	0.87	20	0.74
gambling	12	1.00	0.92	0.96	5	0.50
gambling-adjacent	92	0.72	0.79	0.76	17	0.78
creator community	269	0.78	0.72	0.75	47	0.67
other	412	0.63	0.71	0.67	87	0.57
<i>ST3, macro-F1 0.6199 / 0.5966</i>						
undisclosed advertising	352	0.87	0.86	0.87	74	0.84
inadequate disclosure	611	0.48	0.61	0.54	118	0.45
direct exhortation	304	0.45	0.56	0.50	77	0.54
misleading claim	1277	0.74	0.81	0.78	260	0.78
age-restricted product	59	0.66	0.69	0.68	16	0.69
HFSS food marketing	40	0.42	0.70	0.52	5	0.26
no flag	529	0.57	0.60	0.58	127	0.54
insufficient context	15	0.54	0.47	0.50	7	0.67

Table 4: Per-class results of the system on the two labelled splits under the present-label convention: gold count *n*, precision, recall and F1 out of fold, and gold count and F1 on dev. A class absent from both gold and predictions is dropped from the average (dashes).

gold \ predicted	PG	DC	PS	N	O
PG (physical goods)	1064	38	15	8	0
DC (digital content/services)	32	1009	11	17	0
PS (physical services)	8	23	88	3	0
N (none)	5	4	2	24	0
O (other)	1	1	0	0	0

Table 5: ST1 confusion matrix on the out-of-fold split, gold in rows and predictions in columns.